Malware Analysis Report

RAT.Unknown2.exe

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Executive Summary

Upon detailed analysis, there are no immediate, overt indicators of compromise (IOCs) present within the host system. However, behavioral observation reveals that the malware performs a DNS query for the internal domain: *aaaaaaaaaaaaaaaaaaa.kadusus.local*.

Once the DNS resolution is successful, the malware initiates a remote TCP connection to the resolved IP address. This connection appears to facilitate remote code execution (RCE), allowing an external actor to execute arbitrary commands on the compromised system.

To mitigate this threat, it is recommended to:

- Blacklist the domain aaaaaaaaaaaaaaaaaaa.kadusus.local at the DNS resolver level to prevent resolution.
- Apply egress filtering on firewalls to restrict unauthorized outbound TCP connections, especially to domains resolving to internal .local or unusual FQDNs.
- Monitor and alert unusual DNS queries, particularly domains with excessive or randomized characters.
- Create alerts for outbound TCP connections initiated via cmd.exe or powershell.exe, as this behavior is atypical and commonly associated with malware activity.
- Enable Sysmon Event ID 3 (Network connections) and correlate with Event ID 1 (Process creation) to detect malicious behavior patterns.

Static Analysis

Hashes

MD5	c211704777e168a5151de79dc87ffac7
SHA1	1f688b4872f8a740872b5cb6d58e2b9c7103143a
SHA256	481eae82ac4cd1a9cfadc026a628b18d7b4c54f50385d28c505fbcb3e999b8b0

Data Size

Virtual Size	55,080
Raw Data Size	55,296
Assumption: Malware is unpacked	

Strings

```
\tStartAddressOfRawData

\tEndAddressOfRawData

\tAddressOfIndex

\tAddressOfCallBacks

\tSizeOfZeroFill

 \tCharacteristics

IID_IXMLHttpRequest

CLSID_XMLHTTPRequest

LPORT__GR8mtlZIFUftE9c15JJbgZw

@cmd.exe /c
```

Windows APIs

Enumeration	
GetCurrentProcess	Used to retrieve a handle for the current process.
GetCurrentProcessId	Used to retrieve the process identifier of the specified process.
GetCurrentThreadId	Used to retrieve the thread identifier of the calling thread.
VirtualQuery	Used to retrieve information about a range of pages within the virtual address space of a specified process. This function is commonly used by malware to enumerate an external process.

Injection	
VirtualAlloc	Used by malware to allocate memory as part of process injection.
VirtualProtect	Used by malware to modify memory protection (often to allow write or execution).
CreateProcess	Used to create a process. This function is used by malware in several process injection attacks such as process hollowing.

Helpers	
CreatePipe	Used to create an anonymous pipe and return handles to the read and write ends of the pipe. Could be used to add a sub-process for execution via cmd.
CreateNamedPipe	**NO DATA**
WriteFile	Used to write data to the specified file or input/output (I/O) device.

Evasion	
select	Used to determine the status of one or more sockets to perform synchronous I/O. This function is used by malware for time-based evasion by setting a large timeout number.

Anti-Debugging	
QueryPerformanceFrequency	Used to retrieve the frequency of the performance counter. This function is commonly used by malware for anti-debugging purposes. The malware will measure the time before and after an operation, if the time exceeds taken expected time, the malware will terminate or activate a benign function.

Internet	
send	Used to send data on a connected socket.

recv	Used to receive data from a connected socket or a bound connectionless socket.
socket	Used create a socket that is bound to a specific transport service provider.
closesocket	Used to close an existing socket.
connect	Used to establish a connection to a specified socket.

Unknown	
RtlAddFunctionTable	**NO DATA**
inet_ntop	**NO DATA**
getaddrinfo	**NO DATA**
WSAFDIsSet	**NO DATA**

Dynamic Analysis

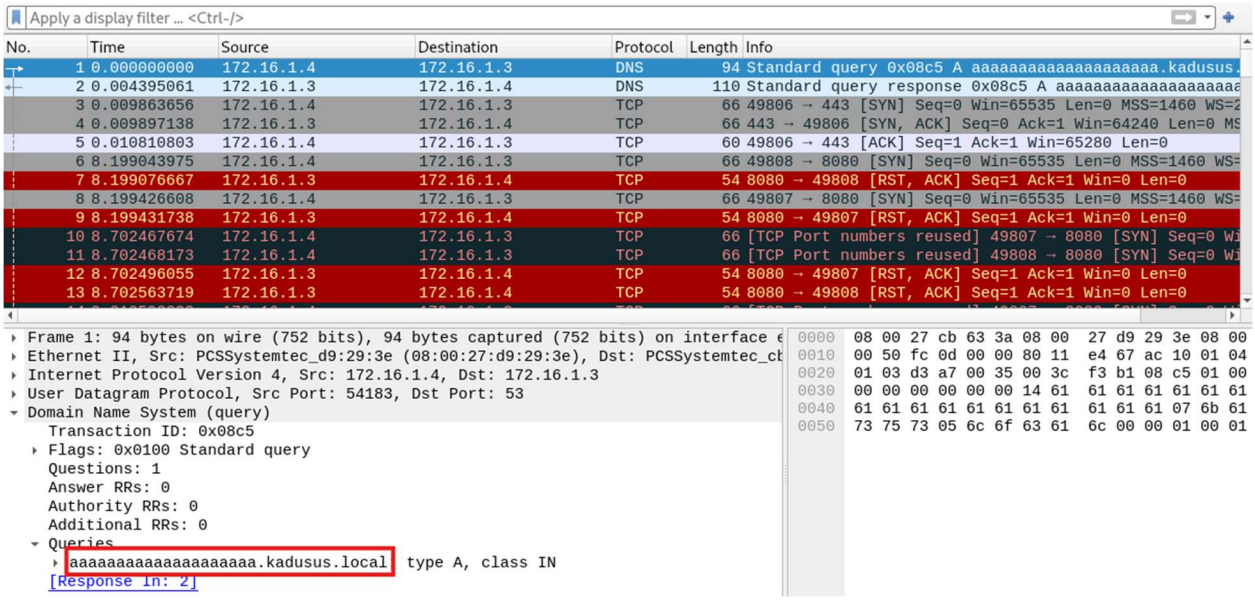
Initial Detonation

Initial detonation does not display any obvious changes to the host

Network-Based Signatures

WireShark

The malware makes a DNS request searching for aaaaaaaaaaaaaaaaaaa.kadusus.local

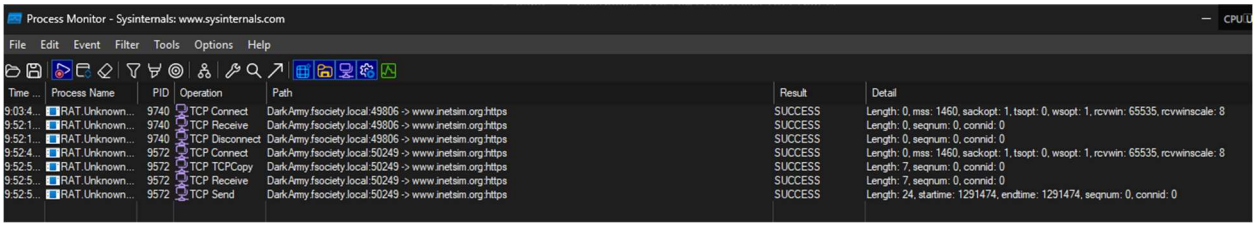


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	172.16.1.4	172.16.1.3	DNS	94	Standard query 0x08c5 A aaaaaaaaaaaaaaaaaaa.kadusus.
2	0.004395061	172.16.1.3	172.16.1.4	DNS	110	Standard query response 0x08c5 A aaaaaaaaaaaaaaaaaaa
3	0.009863656	172.16.1.4	172.16.1.3	TCP	66	49806 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=2
4	0.009897138	172.16.1.3	172.16.1.4	TCP	66	443 → 49806 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MS
5	0.010810803	172.16.1.4	172.16.1.3	TCP	60	49806 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
6	8.199043975	172.16.1.4	172.16.1.3	TCP	66	49808 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=
7	8.199076667	172.16.1.3	172.16.1.4	TCP	54	8080 → 49808 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
8	8.199426608	172.16.1.4	172.16.1.3	TCP	66	49807 → 8080 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=
9	8.199431738	172.16.1.3	172.16.1.4	TCP	54	8080 → 49807 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
10	8.702467674	172.16.1.4	172.16.1.3	TCP	66	[TCP Port numbers reused] 49807 → 8080 [SYN] Seq=0 Wi
11	8.702468173	172.16.1.4	172.16.1.3	TCP	66	[TCP Port numbers reused] 49808 → 8080 [SYN] Seq=0 Wi
12	8.702496055	172.16.1.3	172.16.1.4	TCP	54	8080 → 49807 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
13	8.702563719	172.16.1.3	172.16.1.4	TCP	54	8080 → 49808 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Host-Based Signatures

ProcMon

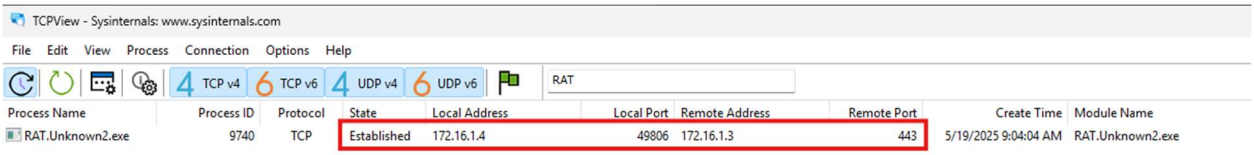
When filtered to show operations that contain TCP and refer to the Process Name RAT.Unknown2.exe, the analyst discovered that the application made TCP connections to a remote host (in this case REMnux) over port 443:



Time ...	Process Name	PID	Operation	Path	Result	Detail
9:03:4...	RAT.Unknown...	9740	TCP Connect	DarkArmy.fsociety.local:49806 -> www.inetsim.org:https	SUCCESS	Length: 0, mss: 1460, sackopt: 1, tsopt: 0, wsopt: 1, rcvwin: 65535, rcvwinscale: 8
9:52:1...	RAT.Unknown...	9740	TCP Receive	DarkArmy.fsociety.local:49806 -> www.inetsim.org:https	SUCCESS	Length: 0, seqnum: 0, connid: 0
9:52:1...	RAT.Unknown...	9740	TCP Disconnect	DarkArmy.fsociety.local:49806 -> www.inetsim.org:https	SUCCESS	Length: 0, seqnum: 0, connid: 0
9:52:4...	RAT.Unknown...	9572	TCP Connect	DarkArmy.fsociety.local:50249 -> www.inetsim.org:https	SUCCESS	Length: 0, mss: 1460, sackopt: 1, tsopt: 0, wsopt: 1, rcvwin: 65535, rcvwinscale: 8
9:52:5...	RAT.Unknown...	9572	TCP TCPCopy	DarkArmy.fsociety.local:50249 -> www.inetsim.org:https	SUCCESS	Length: 7, seqnum: 0, connid: 0
9:52:5...	RAT.Unknown...	9572	TCP Receive	DarkArmy.fsociety.local:50249 -> www.inetsim.org:https	SUCCESS	Length: 7, seqnum: 0, connid: 0
9:52:5...	RAT.Unknown...	9572	TCP Send	DarkArmy.fsociety.local:50249 -> www.inetsim.org:https	SUCCESS	Length: 24, startime: 1291474, endtime: 1291474, seqnum: 0, connid: 0

TCPView

In TCPView the analyst observed that the malware makes a TCP connection over port 443 to the analyst's REMnux box:



Correlation

If we modify the *C:\Windows\System32\drivers\etc\hosts* file to direct traffic intended for the *aaaaaaaaaaaaaaaaaaaa.kadusus.local* domain to the REMnux listener on port 443 the analyst discovered that running the malware creates a reverse TCP connection and granted the remote host code execution capabilities: